

Optimizing Sustainable Farming with AI-Driven Insights

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Business Case

By the year 2050, the global population is expected to reach 9.7 billion of people, this rapid growth poses major obstacles for food security, mostly in terms of efficiency and sustainability (Lal, 2016). This challenge has created the necessity to rely on agriculture innovation; however, agriculture is dealing with growing difficulties recently due to the increasing effects of climate change (Arora, 2019). According to Chen et al. (2016), conventional farming systems have been severely disrupted due to changes in weather patterns, increased variability, and the emergence of new pests and diseases. The traditional agricultural practices that were successful in the past now are facing limitations due to rising environmental and economic pressures (Kumari et al., 2025). Therefore, sustainable agriculture arises as a key approach to counteract food insecurity, focusing on environmental conservation, the enhancement of soil fertility, water conservation, and biodiversity protection.

Business Objectives

Local farmers in Manitoba face unpredictable weather, volatile market prices, and environmental consequences from overusing water and pesticides. This project seeks to optimize farming practices for sustainability and economic gain among these smallholder farmers. The business objectives are:

- Increase crop yields by 10%.
- Reduce water and pesticide use by 15%.

Business Success Criteria

The success of this project will be measured using defined quantitative metrics to ensure objective evaluation.

- **Crop Yield Improvement:** Achieve a 10% increase in average crop yield (tons/ha) compared to historical averages in the dataset's farmer records, calculated from model predictions vs. past yields.

- **Resource Use Reduction:** Reduce water usage by 15% (liters/ha) and pesticide application by 15% (kg/ha) while maintaining or improving yield, assessed via sustainability metrics pre- and post-recommendations.

Data Science

This project tackles a supervised regression problem from a machine learning perspective, aiming to predict crop yields (tons/ha) using sustainable farming inputs (e.g., rainfall, soil moisture, pesticide levels). The goal is to identify patterns that optimize yield and resource use, supporting the business objectives of a 10% yield increase and 15% reduction in water and pesticide use for Manitoba smallholder farmers.

Data Science Goals

- Predictive Model: A regression model forecasting crop yield (tons/ha).
- Processed Dataset: Cleaned data with engineered features (e.g., water per yield).
- Report: This APA-formatted document with model insights.

Data Sources

AI for Sustainable Agriculture Dataset - Kaggle

The data source for this project is “AI for Sustainable Agriculture Dataset” a Kaggle dataset with farmer data (land characteristics, past yields), weather and soil data (rainfall, temperature, soil moisture), and sustainability metrics (water usage, pesticide levels). Assumed adaptable to Manitoba conditions.

- Reference: Suvroo. (2025). *AI for sustainable agriculture dataset* [Data set]. Kaggle. <https://www.kaggle.com/datasets/suvroo/ai-for-sustainable-agriculture-dataset>

Manitoba Weather Data (Supplementary)

If needed, get historical weather records (e.g., precipitation, temperature) from Environment Canada to enhance local relevance if the Kaggle dataset lacks Manitoba-specific climate details.

- Reference: Environment Canada. (2023). *Historical climate data*. <https://climate.weather.gc.ca>

Data Science Success Criteria

- **Model Accuracy:** $R^2 \geq 0.75$ on test data for reliable yield predictions.
- **Predictive Improvement:** 10% Mean Absolute Error (MAE) reduction vs. historical averages (tons/ha).
- **Resource Impact:** 15% reduction in water and pesticide use (liters/kg per ha) in simulations.

Feasibility and Risks

This section assesses the viability of optimizing farming practices for Manitoba farmers using data science across technical, financial, operational, and market perspectives.

Category	Details
Inventory of Resources	Data: AI for Sustainable Agriculture Dataset (Suvroo, 2025). Tools: Laptop, Python (pandas, scikit-learn), Jupyter Lab. Personnel: Solo student effort.
Requirements, Assumptions, and Constraints	Requirements: Kaggle dataset access, basic computing resources. Assumptions: Dataset includes or can adapt to Manitoba-relevant features. Constraints: One-week timeline; no deployment in this phase.
Risks and Contingencies	Technical: Missing data weakens predictions. Contingency: Impute or exclude records. Financial: Minimal cost now; \$5,000-\$10,000 for future rollout. Contingency: Seek grants. Operational: Manitoba farmers may resist change. Contingency: Show cost savings (\$500/ha). Market: Price volatility affects outcomes. Contingency: Focus on sustainable yield gains. Regulatory: Pesticide limits in Manitoba vary. Contingency: Use sustainability metrics.
Costs and Benefits	Costs: Negligible now; \$5,000-\$10,000 for deployment. Benefits: 10% yield increase and 15% resource reduction could save Manitoba farmers. Feasibility is high due to low initial costs and significant potential.

Project Plan

This table outlines steps to achieve data mining goals (model, dataset, report) and business goals (10% yield increase, 15% resource reduction) for Manitoba farmers using Suvroo (2025) and CRISP-DM. Core phases fit within one week, with deployment as a future step.

Phase	Duration	Tasks	Output
Data Understanding	2 days	Explore features (e.g., yield, water use, pesticides)	Data summary
Data Preparation	2 days	Clean data, create features (e.g., water/yield)	Processed dataset
Modeling	2 days	Train regression models for yield and resource optimization	Model with $R^2 \geq 0.75$
Evaluation	2 days	Test accuracy (R^2 , MAE) and resource reduction (15%)	Validated model, report
Deployment	10 days	Share recommendations with Manitoba farmers	Implementation plan

References

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